

Kasdi Merbah University – OUARGLA
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Presented by:

Charaf Eddine MECHALIKH

Rami BOUAFIA

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**IoT Based System for Greenhouses Remote
Monitoring and Climate Control**

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Before the jury:

M.	Abdelkader BENMIR	MAA	President	University Kasdi Merbah -Ouargla
M.	Bachir SAID	MCB	Supervisor	University Kasdi Merbah -Ouargla
M.	Hmida DJEDIAI	MAA	Co-supervisor	University Kasdi Merbah -Ouargla.
Mlle	Chahrazad TOUMI	MAB	Examiner	University Kasdi Merbah -Ouargla

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Dedication

*I dedicate this modest work in the first place to my
parents who supported me to achieve all my objectives
and that they find here all their recognition and my
gratitude.*

To my dearest fiancée

*To my brother, my sister
to my great partner Ramí*

*To my cousin Amel MECHALIKH (may her soul rest in
peace)*

and the whole family

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our teachers*

*at the Department of Computer Science and Information
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their encouragement and the sacrifices they
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To my dear friends, especially my classmates
To all who I love and who love me
To all my colleagues of studies
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Ramí

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General Introduction

Since the 1990s, the internet has become integrated and important in our everyday lives. It has evolved since that time, from simple hypertext pages to internet of content (e-mail, information...), to internet of services where services such as e-commerce were born, to the internet of people or what we know as social media where people communicates with one another, and finally, internet of things (IoT) [33] where things communicate and interact (machine to machine).

While internet changes, the human population is also changing, the population will reach 9.7 billion in 2050, which is 34% higher than today [60]. Growing enough food for a booming world population as the climate changes are cited often as being among the greatest challenges to face humanity [21].

To face this population growth, it's become a priority to increase food production. With the increase of urbanization which is the shifting of people from rural to urban and the growth of urban areas over agricultural lands, and the climate changing (floods, heat waves, and droughts which are occurring with increasing frequency), these two factors affects both crop yields and agricultural lands around the planet which reduces food production and may lead to a food crisis, in this case the greenhouse appears to be the best option as it allows to produce vegetables out of their cultivation seasons in a fully controlled environment, and in small areas, like rooftops greenhouses for example that can increase urban agriculture by making use of the buildings roofs.

Internet of Things (IoT) presents present an enormous opportunity and can solve many problems. Smart Farming is among IoT applications, it helps farmers to increase production with minimal losses and lower costs.

In this project, we will use the latest IoT technology, to get the best of Greenhouse cultivation by reducing waste, and improving agricultural productivity, all this through the greenhouse automation and remote monitoring. To do so, the project was divided into two parts: the part of greenhouses remote monitoring and their climate control, and the part of following plant development and growth.

This work is interested in the first part, and it is divided into three sections. In the first section, we present the state of the art of greenhouse monitoring and IoT. A conception of IoT-based greenhouse monitoring and climate control system is presented in the second section. The last section describes the realization of an automated greenhouse prototype and an implementation of the greenhouse monitoring and climate control system.

1st Part: State of The Art

1.1. Introduction

In order to implement an IoT-based solution for greenhouses monitoring and climate controlling, first, we have to know the greenhouse climate factors that we need to manage, and what the available techniques to control it are, then it is essential to know what IoT is and what can it bring to the greenhouse agriculture.

1.2. Greenhouses Monitoring and Control

Greenhouse environmental control systems owe to the interaction of many environmental variables affecting growth and production. The clear necessity of matching food production to the needs of the world's population requires the accurate knowledge of the factor limiting primary production. Therefore it is essential to identify and understand these environmental factors that limit agricultural production and physiological responses of plants to environmental stresses [46].

1.2.1. Environmental Factors

a. Temperature

Temperature plays an important role in plants growth, Plant growth requires heat. It controls the rate of water and nutrient uptake, photosynthesis, and even cell division. Each plant has different optimum temperature requirements, and different growth stages have different optimums. Seedlings often benefit from warmer temperatures [52], where mature plants tend to have smaller leaves and flowers, paler color, distorted stems and elongated knots [40].

Soil temperatures are more crucial than air temperatures. When soil temperatures are below 7°C, roots work slowly and have a harder time taking up water and nutrients. When the air temperature begins to heat up, wilting can occur even though there is ample moisture in the cold soil because of slow water uptake. Heating the soil, except for seedling flats, may not be worth the effort [52]. In an unheated greenhouse, the main source of heat during the daytime is solar radiation, part of which is stored in the soil. During the night, the energy comes mainly from the soil, in the form of far IR radiation [14].

b. Humidity

"Relative humidity" (RH) is the term used to describe the amount of water vapor present in air expressed as a percentage of the amount needed for saturation at the same temperature [43].

Plants have small openings in their leaves, from which gasses such as CO₂ and oxygen pass. Also, because plants do not use all the water they take, a large amount of water vapor comes out through these openings. This vapor, along with the vapor already in the air, causes high levels of humidity. In a greenhouse that has very little air infiltration from the outside, heat is retained and relative humidity is often very high. Plants grow best at a relative humidity between 45 and 60 percent. Higher humidity often may lead to more diseases [52]. When the humidity in the greenhouse remains less than 45 percent for long periods, plant growth suffers, because leaves lose water faster they can replace it. Otherwise, if the humidity exceeds 80% for any length of time, the risk of leaf diseases increases [40]. Low humidity can happen when much outside venting is going on and it causes plants to wilt [52].

c. Light

All green plants need light so the process of photosynthesis can occur, but the intensity and duration of the needs differ from one species to another. In general, flowering plants need twice as much light as those grown for their leaves. The critical qualities for the light of the greenhouse are how much plants receive (intensity) and how long it lasts. The intensity is determined by the strength of the light source itself and its distance from the plants [40].

When light hits the surface of the leaf, the energy of the sunlight combines with carbon dioxide (CO₂) in the air and water from the soil. By means of photosynthesis, these elements are converted into oxygen and sugar. The oxygen is given off into the air as a component of water vapor, while the sugar is oxidized (burned) in the plant to provide all the energy it needs for growth [52].

d. Carbon Dioxide (CO₂) and Oxygen (O₂)

Carbon dioxide in the air is essential to photosynthesis. Its normal level in the air is about 3% of the air we breathe. When CO₂ is in short supply, plant growth slows. But when the supply of CO₂ increases up to and beyond 300 PPM, up to a certain point plant

growth (including food yield) increases as well. Plants use so much of the CO₂ in the air that in sealed environments like a greenhouse, the level of CO₂ may be depleted from 300 PPM to 100 PPM by noon. This can easily slow plant growth by 60 percent [52].

All living things get the energy they need to live from a chemical reaction called respiration. This process needs glucose as a starting point. In bright light photosynthesis rate is bigger than respiration this mean CO₂ is taken an Oxygen is given out, however in dark there will be only respiration (no photosynthesis) which mean Oxygen is taken and not given out [10].

1.2.2. Existent Greenhouses Environment Control Techniques

There are many used techniques, used to control greenhouse environment, these are some of the popular techniques used to adjust the environmental factors:

a. Light

Supplemental lighting: Most vegetables require at least 8 hours of light per day to produce satisfactorily. In short winter days, or if in very cloudy areas, supplemental lighting will increase the production [52].

Increasing reflecting areas: Reflected light is a powerful concept that has enabled many urban gardens to succeed. Reflected light is indirect sunlight that is bounced off reflective or light-colored surfaces, and it can still bring a lot of energy to the plants. For growing fruiting vegetables such as peppers or tomatoes, in a garden which only gets 3 or 4 good hours of sunlight per day, reflected light might make up the difference [17].

Shading: when too much light is in the greenhouse [27].

b. Carbon dioxide

CO₂enrichment (Dry Ice method): Commercial greenhouses commonly use CO₂ generators to maximize production. While designing a CO₂ system, yields will only increase if CO₂ is the 'limiting factor'. This means that if all other variables are not optimal (light, fertilizer, temp/humidity, pH, etc.) the benefits of increased CO₂ levels will not be achieved [23].

CO₂ventilation: The limited air movement hinders the supply of CO₂ to the stomata of the leaves for photosynthesis [14], creating an envelope of CO₂ deficiency around leaves

which result in a decrease of photosynthesis rate [52]. The air movement affects the plant growth, altering the energy transfers, the transpiration, and the CO₂ absorption so that the leaf size is affected, as well as the stem growth and the yield [14]. The rate of photosynthesis can be increased by 40 percent if the wind velocity increases from 10 to 100 centimeters per second [52].

c. Humidity

High humidity: Ventilating to the outside whenever the outside temperature is warm enough, ventilate the moist air to the outside. This can also be done when the greenhouse is running hot, even if it is not warm outside [52].

Low Humidity: Low Humidity may be increased by simply watering the floor and by using swamp cooler type air conditioners. Another way is to use a misting system to spray mist into the air. This will help cool the greenhouse while adding moisture to the air [52].

d. Temperature

Cooling

- **Evaporative cooling:** The principle is simple. As the exhaust fans blow air out of one end of the greenhouse, they draw in moist air from the other end. As the moist air moves through the greenhouse, some of the water vaporizes, absorbing heat in the process. Moisture is supplied at the end opposite the fans with a system that drips water through an absorbent material such as cellulose or a synthetic fiber (called “cool pads”). All incoming air passes through this wet fiber (**Fig. 1**) [53].

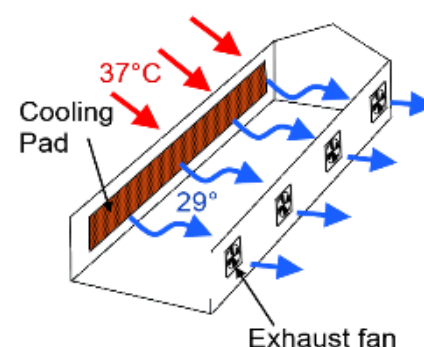


Fig. 1: Evaporative cooling operation.

- **Shade cloth:** The advantage of an internal shade curtain is that it can be pulled back on cloudy days and spread out on sunny days as needed, either manually or with small motors. The disadvantage of using an internal shade cloth is that light is converted to heat inside the greenhouse, although plants receive less light, they don't get the same reduction in heat. If the shade cloth is used outside the greenhouse, sunlight will be converted to heat before entering the greenhouse [53].

- **Passive cooling:** Air movement is just as important as ventilation. As air warms it rises, opening the roof (or using an extractor fan on the top of the greenhouse) while opening the walls will encourage the warm air to leave. [16] (**Fig. 2**).

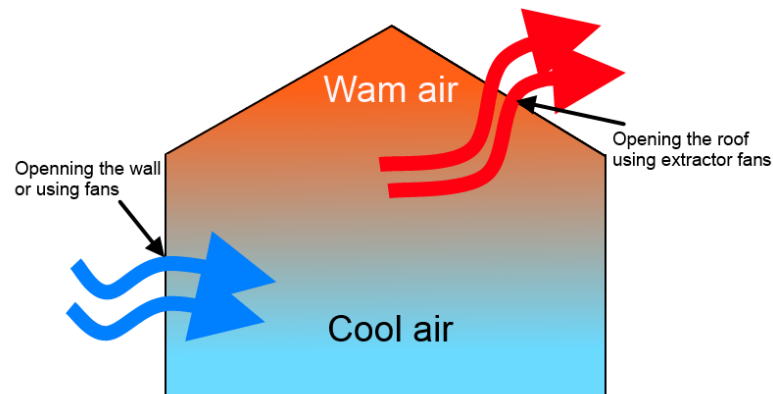


Fig.2: Passive cooling operation.

Heating

- **Electric greenhouse heaters:** These are great if the greenhouse is having a power supply and because they can be thermostatically controlled, heat is delivered only when required so running costs need not be as high [2].
- **Gas heating:** The heat came from a normal mains gas central heating border. Wonderful system, but very expensive to install. The running costs were quite modest though [2].

1.3. Internet of things and Greenhouses

The Internet of Things (IoT) paradigm is based on intelligent and self-configuring nodes (things) interconnected in a dynamic and global network infrastructure [13]. These things can be computing devices, mechanical and digital machines, objects, animals or people that are provided with unique identifiers and the ability to transfer data over a network without requiring human-to-human or human-to-computer interaction [58].

1.3.1. Concepts

The cloud computing: Cloud computing is an infrastructure in which computing power and storage are managed by remote servers to which users connect via a secure Internet connection using a computing device (PC, smartphone...) which become an access point to view data or run application that is hosted on these servers [24]. It allows users to consume a compute resource, such as a virtual machine, storage or an application, as a utility rather than having to build and maintain computing infrastructures in house [51].

IoT platform: An Internet of Things (IoT) platform is the support software that connects edge hardware, access points, and data networks to other parts of the value chain (generally the end-user applications). IoT platforms typically handle ongoing management tasks and data visualization, which allow users to automate their environment [48].

1.3.2. Uses of IoT in the Greenhouses

IoT is used in many different domains, from environment protection (forests trees protecting, air monitoring...), energy (wind turbines, electricity meters...), public safety (bridges, buildings, vehicles...), health (baby monitors, elderly monitors...), to agriculture [15], especially in greenhouses where it can be used for:

a. Security

There are many ways where IoT can be used for security: infrared radiation for fire or intrusion detection [13], Video surveillance [29], equipment tracking [35]...

b. Environmental monitoring

Some applications can be related to the continuous and long-term monitoring of gas concentration in air, soil humidity, lighting conditions [13].

c. Climate control

Temperature, humidity and soil moisture can be monitored through various sensors. These can then be linked to systems to trigger alerts or automate processes such as water and air control [22].

d. Crop monitoring

Rosphere is a farm robot built by The Technical University of Madrid, which is armed with sensors and can potentially monitor every single stalk in a field. These robots can be configured to communicate to one another over a network. The data can

be collectively used to build information sets such as crop yield maps, and further linked to information such as current crop prices [22].

1.3.3. Current IoT Solutions for the Greenhouses

Kaa: The Kaa open-source IoT Platform. By tying together different sensors, connected devices, and farming facilities, it allows farmers and growers, to remote monitoring climate, crop, and equipment, it also provides predictive analytics for crops [35].

GetSenso: Greenhouse monitoring with wireless sensor technology and IOT offers a low-cost monitoring parameter within a greenhouse. By placing different sensors like light sensor, humidity sensor etc. farmer can easily monitor it from a remote location. If the farmer is unable to visit the greenhouse he can initiate an automation process like watering or provide shade when too much of light is present [27].

With GetSenso application, the user can monitor the greenhouse parameters from its device location through a remote desktop computer, a mobile phone or tablet. Using GetSenso cloud application logging and monitoring of sensor data takes place in real time. The application also provides input to set the automation values of the greenhouse (**Fig. 3**) [27].

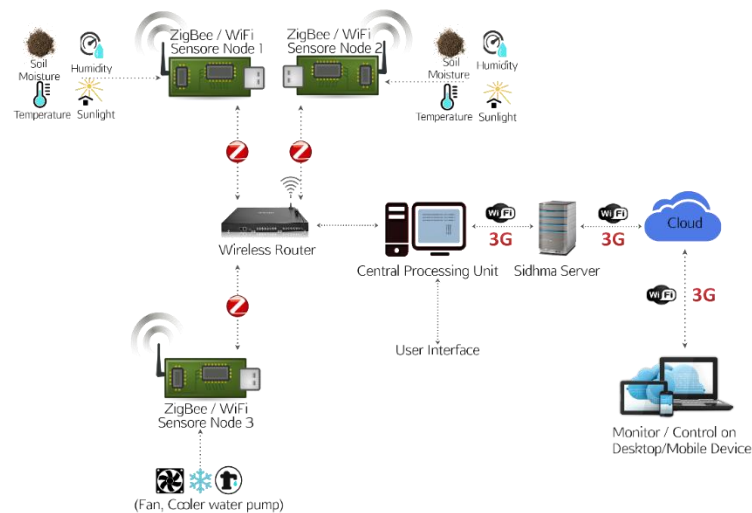


Fig. 3: GetSenso, How it works [27].

Bitponics: Is an open-source IoT gardening project that offers to growers and farmers plan reminders which notify when they should take action, real-time sensor readings, also it

offers the ability to automatically turn connected devices on & off, to manage their garden and to track progress over time from any web browser (**Fig. 4**)[11].



Fig. 4: Bitponics, how it works [11].

Plantlink: Plantlink is a soil moisture sensor that will send to the farmer a watering reminder (push, email, text) when it's most convenient. This means watering precisely when the plant needs it. If the farmer wasn't there, plant link sensors can notify the plant link valve to automatically watering the plant (**Fig. 5**) [45].



Fig. 5: Plantlink sensor and valve [45].

HarvestGeek: An open-source, wireless, Internet of Things Greenhouse monitoring and automation system for your farm or garden (**Fig. 6**). Once configured it will automatically notify you based on the information being received from HarvestBot. You can choose to

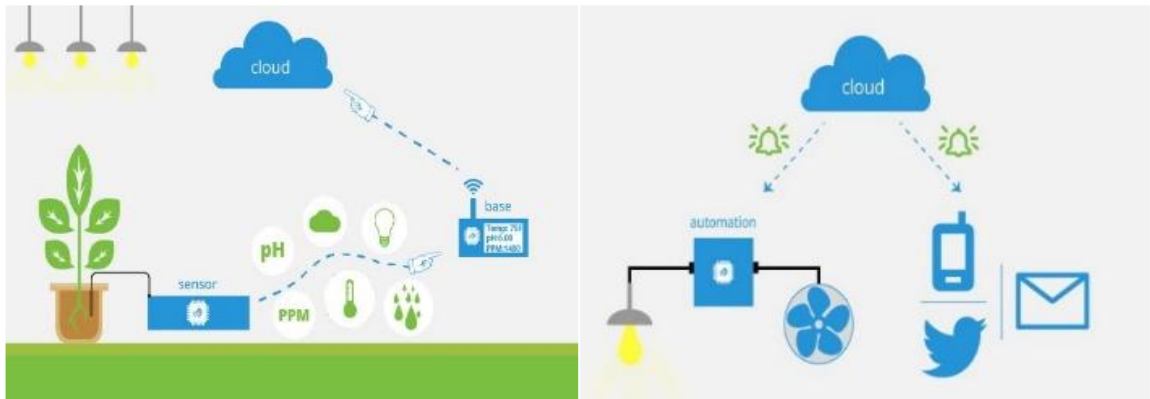


Fig. 6: HarvestGeek, monitoring and automation [37].

receive notifications via SMS, Facebook, Twitter, or Email should your plants reach an unhealthy situation that calls for your attention. You can also configure the Automation Station to control equipment [37].

1.4. Conclusion

It is always important to make a preliminary study when starting any project, in order to start on a strong basis.

Now after defining the important sides of our problem which are the greenhouse climate factors, the techniques to control them, and the IoT uses in greenhouse, we're ready to use them in the next phase which involves analyzing and designing the system.

2nd Part: Conception

2.1. Introduction

The project aims to automate and to remote monitoring greenhouses using IoT technology. So the farmer will not need to go every time to the greenhouses and check their status.

This part is about the conception which is the method of analysis and design that aims to formalize the preliminary stages of the development of a system in order to make this development more faithful and answers the needs. To do this we will start with greenhouse automation scenario by using the techniques from the previous part, then it is important to talk about risks and failures detection and how to notify the user if something wrong occurs, and the remote monitoring scenario which uses the previous two tasks.

2.2. Architecture

Network Topology is the schematic description of a network arrangement, connecting various nodes (sender and receiver) through lines of connection [54]. The three main topologies are: Bus, Ring and star topology, each of them has its advantages and disadvantages, the following table (**Tab. 1**), represent a comparison between them:

Topology	Installation	Cable/node failure	troubleshooting	Performance
Bus	Easy to setup and modify	Whole network fails	fairly easy	medium
Ring	Hard to modify	Whole network fails	difficult	slow
Star	Easy to setup and modify	Only that node is affected	easy	fast

Tab. 1: A comparison between network topologies [36].

As a conclusion, the star topology is the best one to use in this case, in order to connect the different greenhouses (**Fig. 8**) and to connect the different devices that a greenhouse has (**Fig. 7**), and these devices are:

a. Sensors:

A sensor is a device that detects and responds to some type of input from the physical environment. The specific input could be light, heat, motion, moisture, pressure, or any one of a great number of other environmental phenomena. The output is generally a signal that is converted to human-readable display [57]. These sensors send their output to microcontrollers.

b. Actuators:

An actuator is a device consisting of a piece of machinery that has moving parts that perform some function [61].

c. Microcontrollers:

A microcontroller is a self-contained system with peripherals, memory and a processor that can be used as an embedded system. Some of them are more sophisticated, while others have minimal requirements for memory and programming length and a low software complexity. Input and output devices include solenoids, LCD displays, relays, switches and sensors for data like humidity, temperature or light level, amongst others [25].

These microcontrollers send their output to a device that reads their output and process it. In the greenhouse, microcontrollers read sensors data and send them to the concentrator of that greenhouse, or they turn on/ off actuators, depending on the received command.

d. The concentrator:

A concentrator, in this case, is a credit card sized computer that link between the greenhouse devices (sensors and actuators) and the gateway.

Among the advantages:

- Collecting all greenhouse data.
- It represents the greenhouse on the gateway side (it will send all greenhouse data to the gateway, so the gateway will see it as a greenhouse).

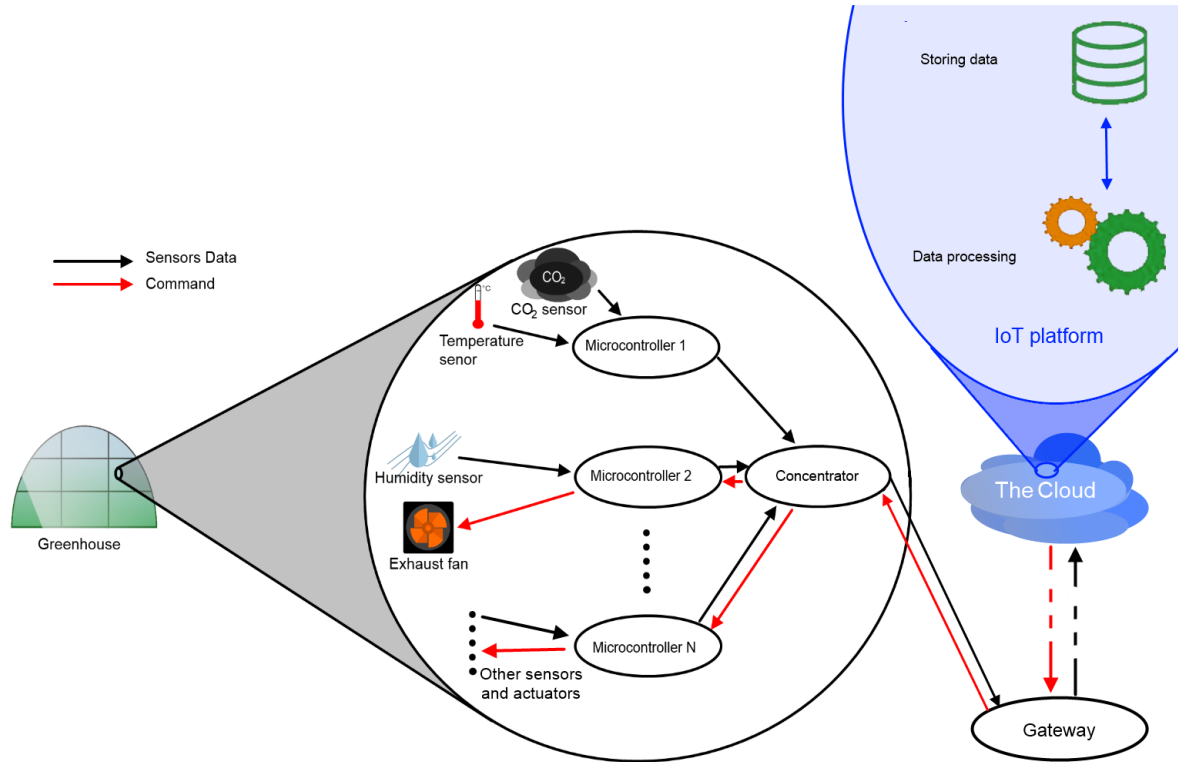


Fig. 7: Greenhouse content.

e. Gateway:

An internet of things gateway allows a device to report data using its sensors to a remote location [55].

Every one of these greenhouses is connected to the same gateway forming a star network (**Fig. 8**), this gateway collects the data from all the greenhouses, filters it (depending on message priority) and sends it periodically to the cloud, which leads to a low cost as IoT cloud platforms use messages count as a payment unit (1 message instead of N messages, where N is greenhouses count).

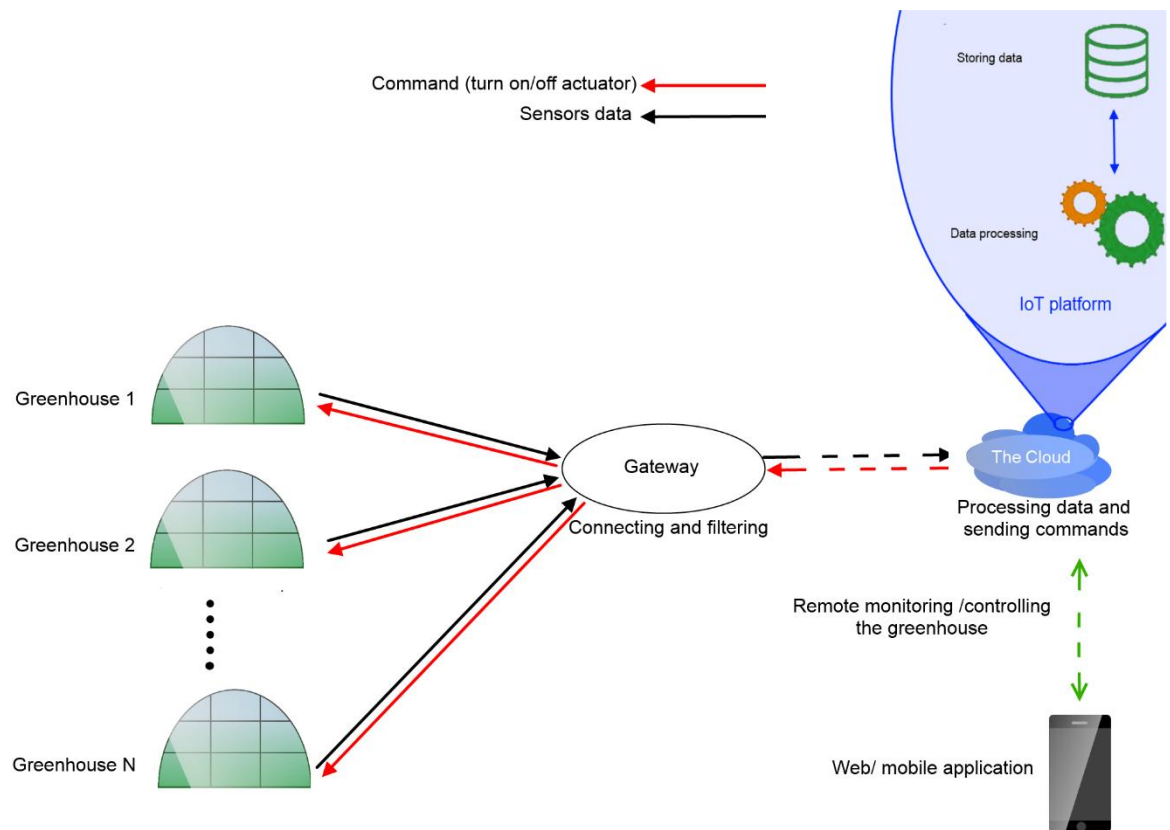


Fig. 8: System architecture.

Since the project is about greenhouses remote monitoring and climate control, it can be divided into three essential parts: greenhouses automation, remote monitoring and controlling, and failures detection.

2.3. Greenhouses Automation

Automation is the linking of disparate systems and software in such a way that they become self-acting or self-regulating [56]. In this case it is the linking of sensors and actuators to get self-acting greenhouses.

2.3.1. Environmental Factors and Possible Actions

Every environmental factor has a max and min threshold that it has to stay in between, the following table (**Tab.2**) describes the possible actions to do, to adjust a factor when it crosses a threshold.

Factor	Condition	Possible actions	Actuators
Temperature	> Max value (very hot)	Evaporative cooling, shading, opening the roof.	Evaporative cooler, Shade cloth, the roof and the wall or fans.
	< Min value (very cold)	Heating by electric or gas heater	Heater
Humidity	> Max value (High humidity)	Ventilating to the outside	Exhaust fan
	< Min value (very dry)	Watering the floor, spraying mist, Evaporative cooling.	Misting system
Light	Intensity > Max value Or Exposure duration > the time needed	Shading.	Shade cloth
	Intensity < Min value (Night, cloudy day...) Or Exposure duration < the time needed	Supplemental lighting	Bulbs
Carbon dioxide	< Min value	CO2 enrichment	Dry ice + ventilating
Oxygen	< Min value	Aerate the greenhouse	Open walls and roof, or fans.
Wind	High wind	Close/don't open the roof/ wall	The roof/wall

Tab. 2: Factors values and the possible actions.

2.3.2. Automation Scenario

To automate the greenhouses using IoT, the sensors data will be transmitted through a set of different devices, to the cloud where it gets processed, if a measurement crossed the threshold, the cloud sends a command to the specified greenhouse in order to perform the suitable action. The diagram below (**Fig. 9**) describes greenhouses automation scenario:

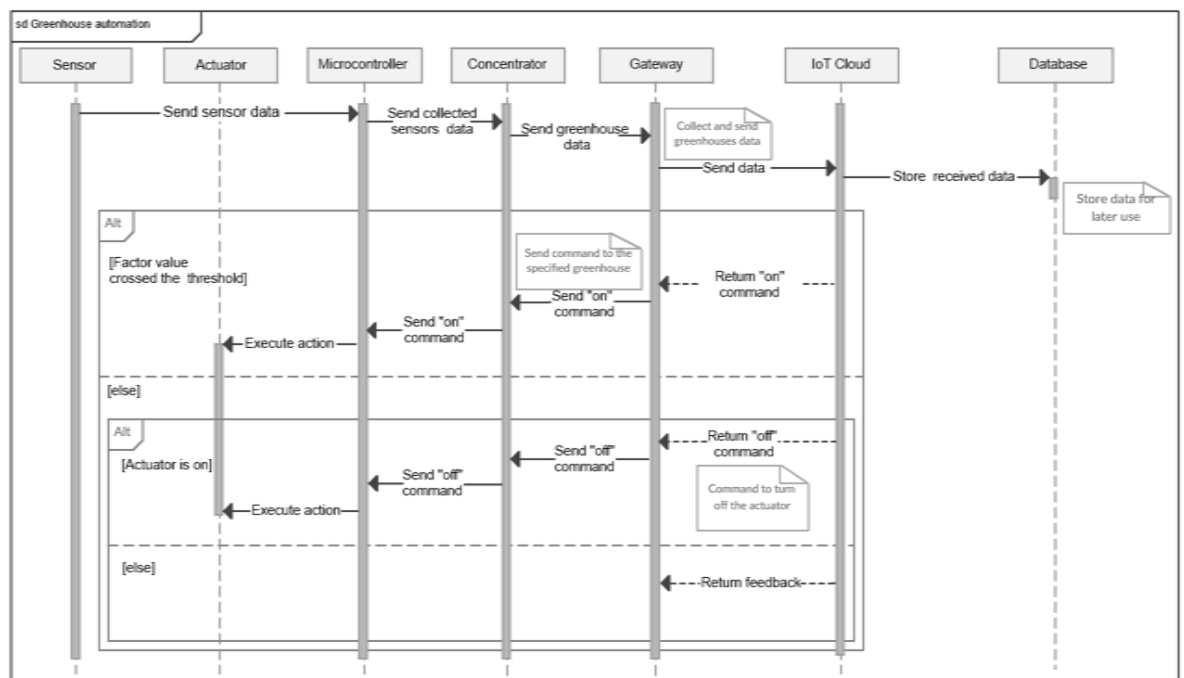


Fig. 9: Greenhouses automation sequence diagram.

2.4. Risks and Failures Detection

The idea here is to notify the farmer when a problem occurs in order to repair it. There are several failures that the system can face, from which we can find:

- Sensor failure: wrong output, this can damage the entire greenhouse.
- Actuator failures.
- Loosing connection with the cloud: this can occur often.

2.4.1. Failures and Actions

The following table (**Tab. 3**) shows the failures, how to detect them, and the best action to do:

Failure	Detection	Actions
Actuators not working properly	If an actuator is running for a known period of time, but the sensor results are not as expected. i.e.: cooling system is running for 30min but temperature didn't decrease.	Send an error message to the user: "the actuator <actuator id> is not working properly/damaged" Or "the cooling, lighting or heating... system is not working properly"
Sensors not working properly	Using 3 sensors of same nature and comparing their results, if the results aren't same then a sensor is not working properly or damaged. i.e.: two sensors give 28°C while the third one gives 32°C, so the third one don't work properly.	Send an error message to the user: "The sensor <sensor id> is not working properly/damaged".
No internet connection	If the cloud don't receive a message for a known period of time.	Send an error message to the user/farmer "No connection between Cloud and greenhouses".
	If the gateway can't send messages to the cloud.	Using the gateway instead of the cloud until connecting again to internet.
Device not working (microcontroller ...)	No messages coming from it.	Send a message to the user/farmer "The device <device id> not working".

Tab. 3: Failures detection.

2.4.2. Failures Detection Scenario

The sequence diagram below represents the failures detection scenario (**Fig. 10**):

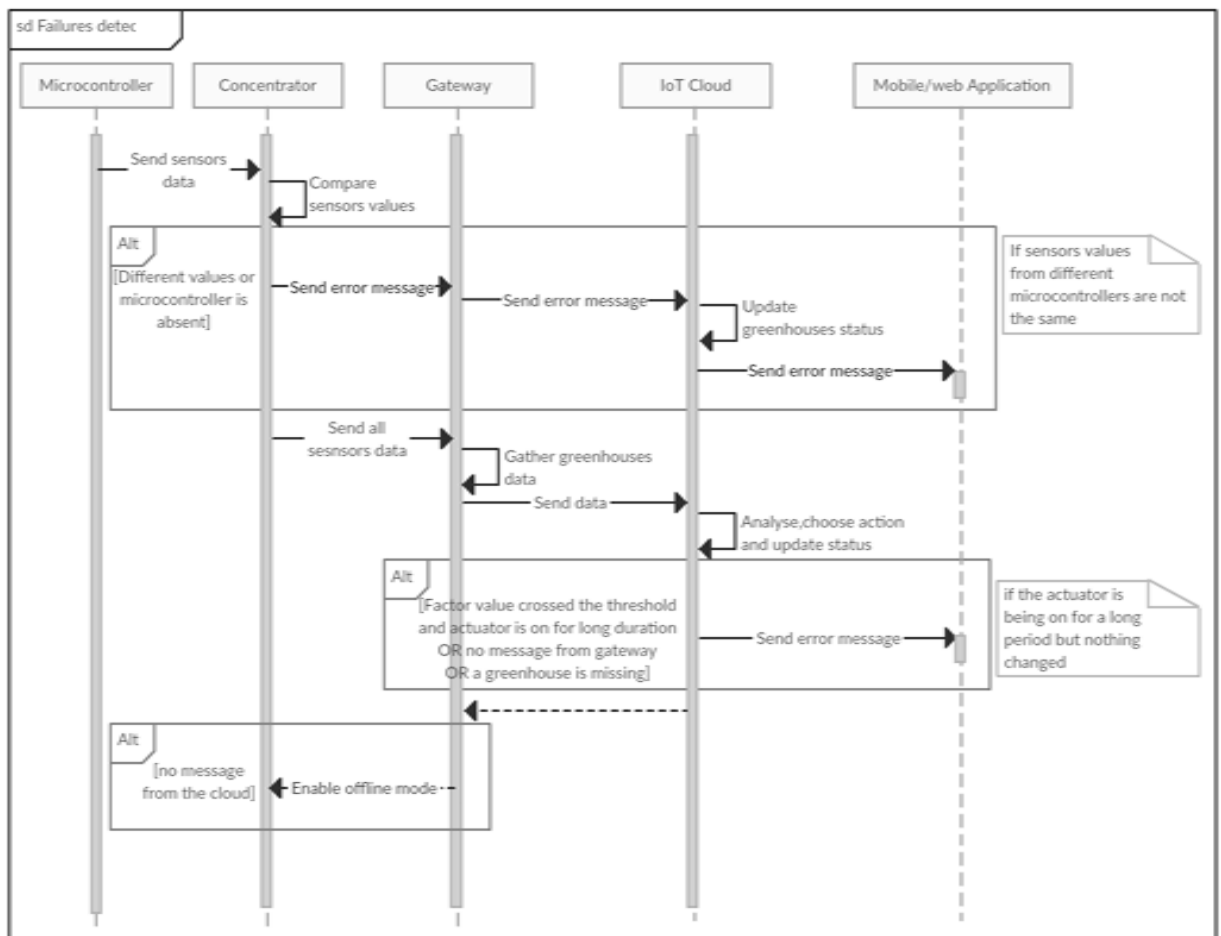


Fig. 10: Failures detection sequence diagram.

2.5. Remote Monitoring and Controlling

The idea behind the remote monitoring and controlling is to allow the farmer to consult the greenhouse status or history, and controlling it from anywhere with IoT, using a mobile application or a website.

2.5.1. Application Tasks

By using the application, the user can monitor the greenhouse (view the environmental factors current values and history, actuators state, and weather information), perform a manual action (turn on or off an actuator), and view the notifications, the following diagram describe the use case diagram of the application (**Fig. 11**).

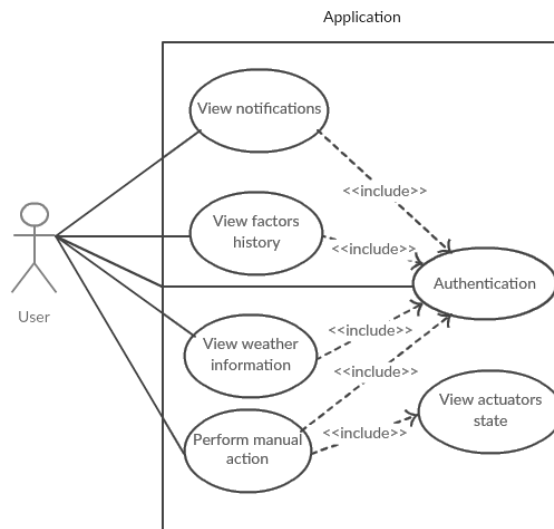


Fig. 11: Application use case diagram.

2.5.2. Application Scenario

Once authentication is successfully done, the application gets the data history from the cloud database and displays it as a chart.

When the user performs a manual action (enable/disable actuator of his choice), the application will send the command to the cloud, which will process it and send it to the gateway, then the gateway sends it to the specified greenhouse. The next sequence diagram shows the remote monitoring and controlling scenario (**Fig. 12**).

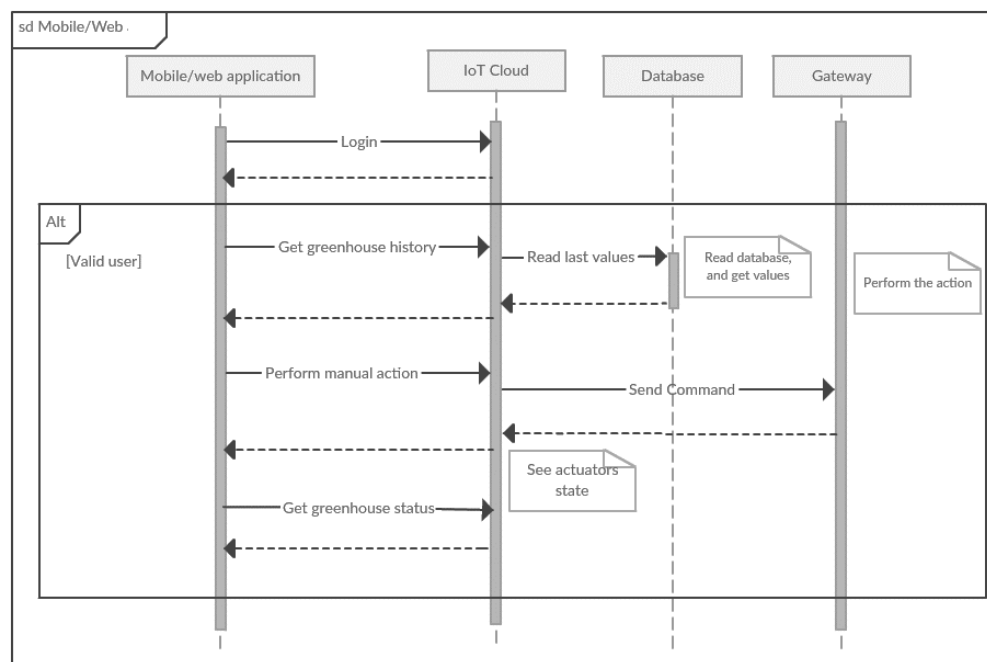


Fig. 12: Greenhouse remote monitoring and controlling sequence diagram.

The following figure represent the activity diagram of the application (**Fig. 13**):

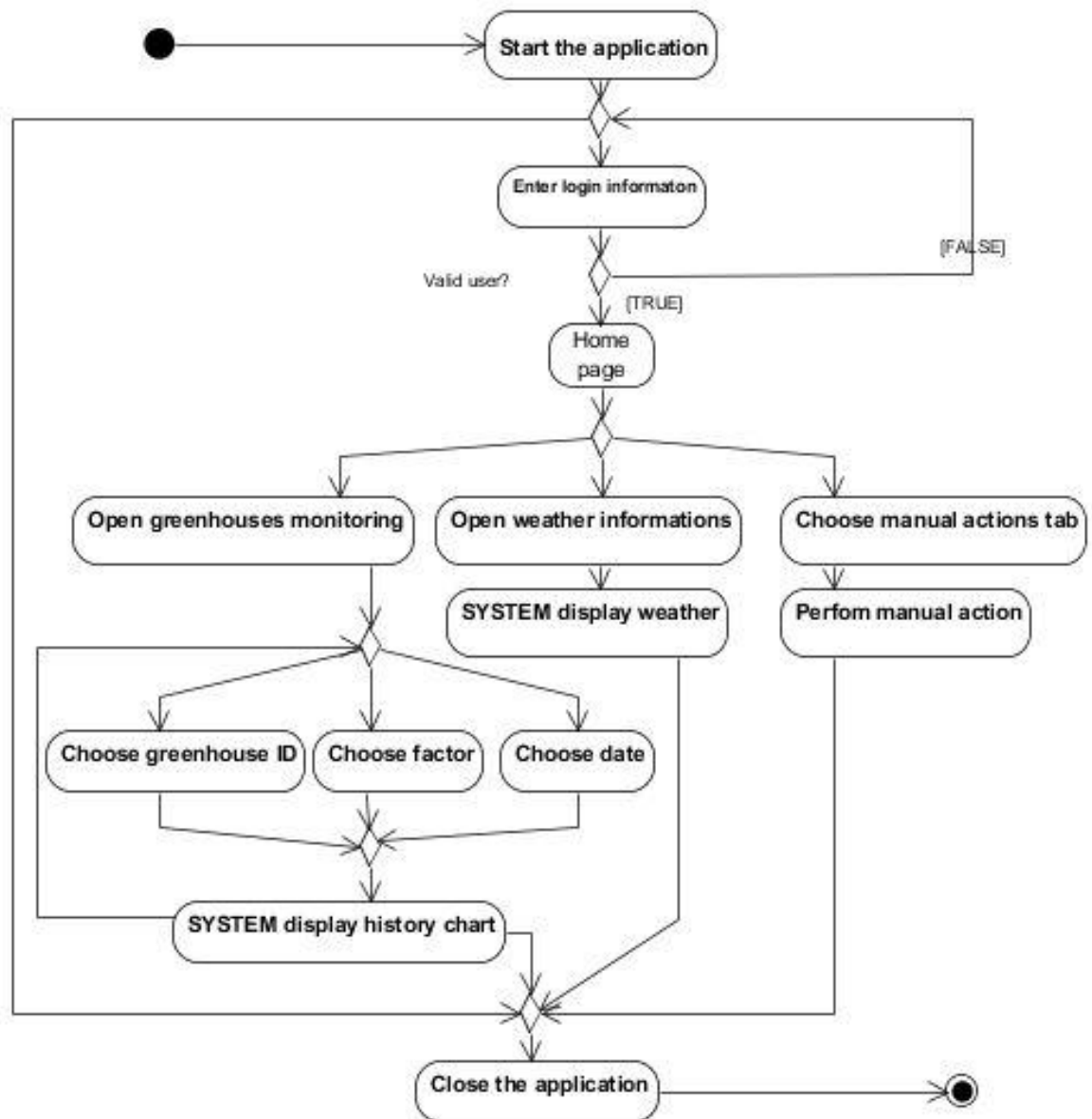


Fig. 13: Application activity diagram.

2.5.3. Designing The User Interface (mobile and web applications)

a. The mobile application

The mobile application has to offer:

- Real-time monitoring of the environmental factors of a greenhouse.
- Real-time weather information (the weather outside the greenhouse).
- The ability of seeing the running actuators and manual controlling them.

Designing the mobile application

Because the application has three major goals, its user interface should have three tabs each one for a specific goal. The following wireframe (**Fig. 14**) represent the result and explains the different parts that the application will have.

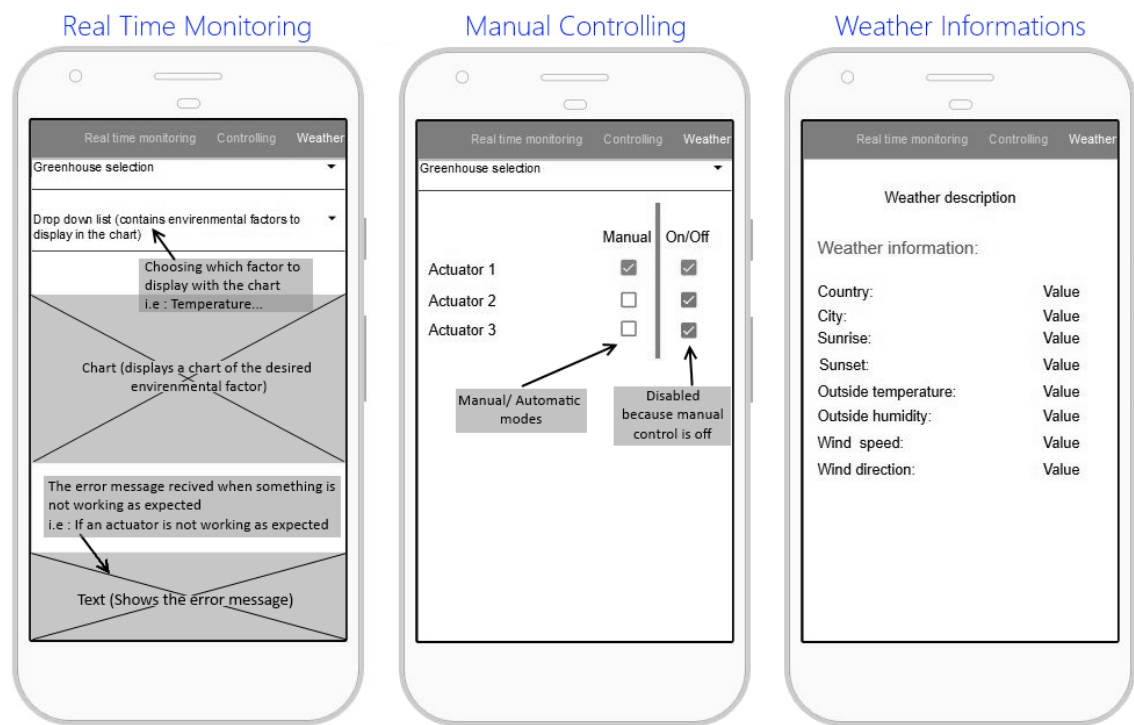


Fig. 14: Mobile application wireframe.

b. The web application

Like the mobile application, the web application should offer a remote monitoring ability, information about the weather outside the greenhouses, display the running actuators and the ability to control them manually. This wireframe (**Fig. 15**) shows its functionalities:

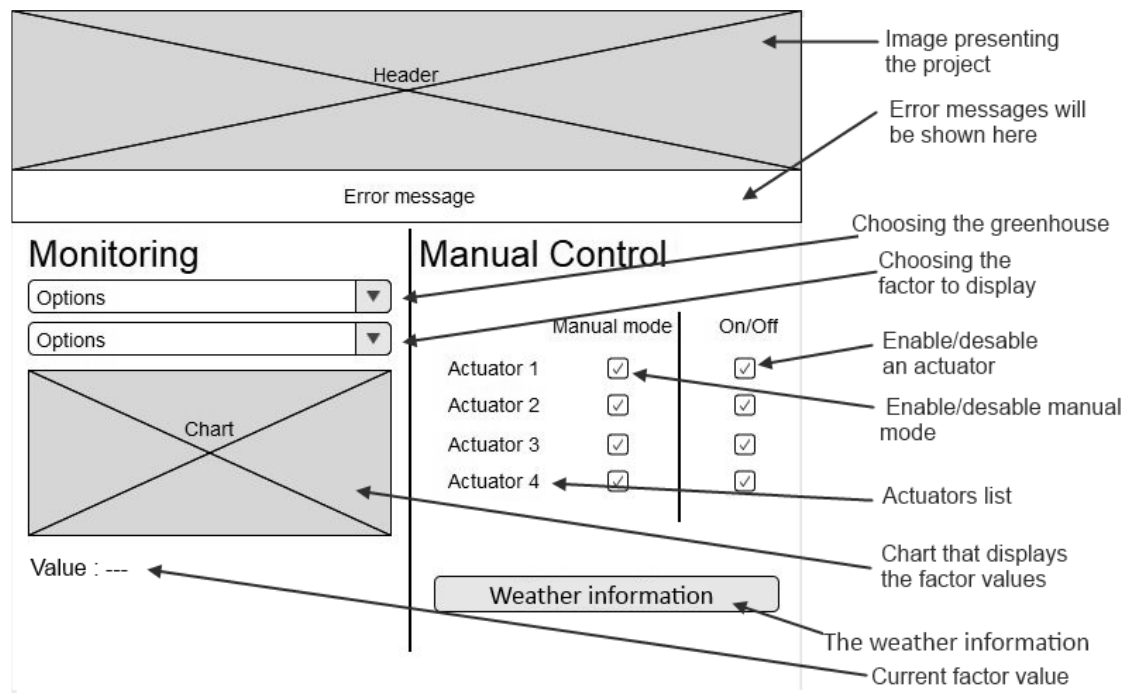


Fig. 15: Web application wireframe.

2.6. Conclusion

This part was about defining the architecture, the scenarios, and the work plan. With the information that we had from the first part, now we can start the next phase, which is about implementing and building a running product from this conception phase.

3rd Part: Implementation

3.1. Introduction

Implementation is the process of creating a finished product from a conception document, it involves the analysis of requirements, installation, configuration, execution, testing and more, depending on the nature of the project.

In this part we will talk about the implementation of the solution which includes the plant we worked on, the chosen IoT platform, the software and the hardware we used, and the node-red configuration.

3.2. Choosing The Plant

Since every plant has its specific needs, the solution will depend on the plant species, so it is important to select one to work on:

Selected plant species: Sweet pepper.

Botanical name: *Capsicum annuum*.

Family: *Solanaceae*.

Growing conditions:

- Temperature between 20°C and 30°C during day, and between 15°C and 20°C during night [31].
- Humidity between 60% and 70% [31].
- It requires 6-8 hours a day of direct sunlight [12], where the light must be superior of 9000 lux [9].



Fig. 16: Growing sweet pepper in the greenhouse.

3.3. Hardware

To implement the solution, we will use these following devices:

- a. Arduino Uno R3 (as microcontroller): Arduino Uno R3 is the third version of Arduino Uno the board (see **Fig. 17**) [26]. The Arduino Uno line, is the best choice for getting started and prototyping [26][5]. However for advanced users the compact



Fig. 17: Arduino Uno.

version called “Arduino Nano” will be the best option [4] (see **Fig. 18**).

- b. Sensors: Humidity and temperature sensor (DHT11), light sensor (Mini Photocell SEN-09088), CO2 Meter and Oxygen sensor.

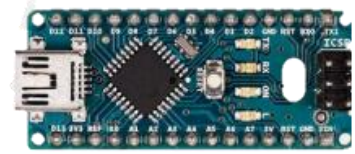


Fig. 18: Arduino Nano.

- c. Actuators: Bulbs (for supplemental lighting), heaters, misting system, retractable roof, walls, fans...

- d. Raspberry pi 3 model B (a credit card sized computer used in every greenhouse to gather the data of sensors (concentrator **Fig. 7**) or used as a gateway between the cloud and the greenhouses **Fig. 8**).

The raspberry pi 3 model (see **Fig. 19**) b comes with [47]:

- A 1.2GHz 64-bit quad-core ARMv8 CPU
- 1GB of RAM
- 4 USB ports
- 802.11n Wireless LAN
- Bluetooth 4.1
- Bluetooth Low Energy (BLE)
- HDMI port...and more.



Fig. 19: Raspberry pi 3 model B [47].

3.4. Choosing IoT Platform

Nowadays, there are various IoT platforms available and can be used for developing an IoT solution. In this section, we have covered three popular platforms that are widely used for IoT solution building.

3.4.1. IBM Bluemix

IBM Bluemix is a platform as a service (PaaS) cloud which is developed by IBM. It supports programming languages like java, php, Python, Node.js, Go and many more. Integrated DevOps allows to build, run, deploy as well as to manage applications over IBM Bluemix cloud [59].

Cloud Foundry is an open-standard, open-source PaaS framework and community, funded and managed by IBM and other vendors. It runs on a broad range of cloud infrastructure platforms, including OpenStack, VMware, and IBM SoftLayer. It also supports a wide range of application programming language runtimes and frameworks, including Java, Ruby, JavaScript, and Python (see **Fig.20**). Cloud Foundry has a large, vibrant ecosystem of partners, vendors, and users, enabling enterprises to develop cloud-centric applications based on its open standard without fear of vendor lock-in [30].

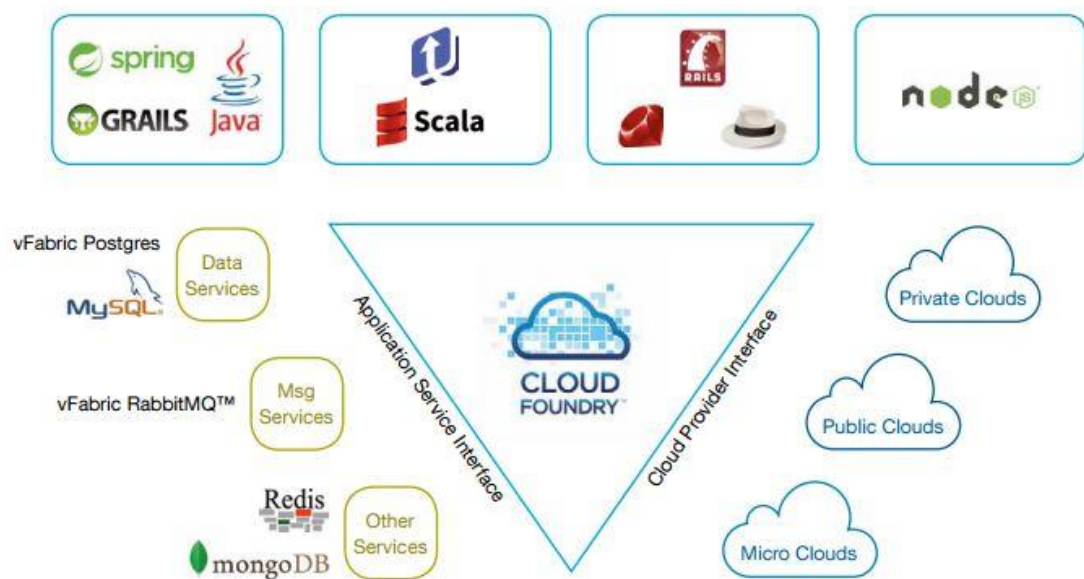


Fig. 20: Cloud Foundry [30].

3.4.2. Microsoft Azure

Microsoft Azure is a growing collection of integrated cloud services that developers and IT professionals use to build, deploy, and manage applications through our global network of datacenters. With Azure, you get the freedom to build and deploy wherever you want, using the tools, applications, and frameworks of your choice [8].

3.4.3. Amazon Web Services

AWS IoT is a managed cloud platform that lets connected devices easily and securely interact with cloud applications and other devices. AWS IoT can support billions of devices and trillions of messages and can process and route those messages to AWS endpoints and to other devices reliably and securely. With AWS IoT, your applications can

keep track of and communicate with all your devices, all the time, even when they aren't connected [3].

3.4.4. Comparison

The following table (**Tab. 4**) shows a comparison between the three well known IoT platforms mentioned earlier [3][7][19]:

	Microsoft Azure	Amazon AWS IoT	IBM Bluemix
Protocols	HTTP, AMQP, MQTT and custom protocols (using protocol gateway project)	HTTP, MQTT, WebSocket	HTTP, MQTT, Sockets/ WebSocket
SDK/languages	.Net and UWP, Java, C, NodeJS	C, NodeJS, JavaScript and Arduino programming languages	Java, C#, php, C, Go, Swift, Ruby Python, NodeJS and more.
Authentication	Per-device with SAS token	X.509 certificate client authentication, IAM service, Cognito service	Per-device with token
Pricing	Free for up to 8,000 messages per month or \$50 per month for 400,000 messages.	\$5 per million messages	30 day trial for free, 1\$ approximately for every 1Gb of messages (400,000 messages can cost from 0.29\$ to 49,90\$ depending on messages size)[32]
Registration	Credit card required	Credit card required	No credit card

Tab. 4: A comparison between three of well-known IoT platforms.

For the implementation of our system we have chosen IBM Bluemix IoT platform since it offers a miscellaneous service catalogue and a 30 day trial which is ideal for the application testing and also a very good pricing as our estimated message size will be around 1Kb.

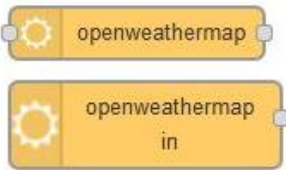
3.5. Development Tools

The development tools needed in order to implement the solution are:

- **Arduino IDE:** The open-source Arduino Software (IDE) makes it easy to write code and upload it to the board. It runs on Windows, Mac OS X, and Linux. It can be used with any Arduino board [6]. The official IDE for programming the Arduino Uno.
- **Android Studio:** for building the mobile application (for Android OS) in Java programming language.
- **Note-Pad++:** recommended to write HTML, CSS and JavaScript web application codes.
- **Node-RED:** Node-RED is a programming tool for wiring together hardware devices, APIs and online services in new and interesting ways. It provides a browser-based editor that makes it easy to wire together flows using the wide range of nodes in the palette that can be deployed to its runtime in a single-click [42], it allows to write functions in JavaScript.

Additional nodes to install in the IoT cloud Node-RED:

Node	Package name	Description	Graphic representation
Suncalc	Node-RED-node-suncalc	A node that provides a signal at sunrise/sunset	
Splitter	Node-RED-contrib-splitter	A node that splits an array and returns its items one by one	

Openweatherm ap	Node-RED-node- openweathermap	A node that gets the weather report and forecast from OpenWeatherMap.	
--------------------	----------------------------------	--	---

Tab. 5: Additional nodes to install in the Cloud Node-RED.

3.6. Communication Modes

There is a multitude of wireless standards that address high data transport. Among them we find the following standards:

a. ZigBee (IEEE 802.15.4)

ZigBee is a LP-WPAN (Low Power-Wireless Personal Area Network) a wireless communication protocol designed by the ZigBee Alliance based on the IEEE 802.15.4 standard that defines the PHY (Physical) and Medium Access Control (MAC) layers [38].



Fig. 21: ZigBee logo.

b. Wi-Fi (IEEE 802.11)

Wi-Fi is technically an industry term that represents a type of wireless local area network (LAN) protocol based on the 802.11 IEEE network standard [44].



Fig. 22: Wi-Fi logo.

c. Bluetooth (IEEE 802.15.1)

Bluetooth is a wireless personal area network (WPAN) technology that is a low-range wireless network technology. It allows devices to be connected to each other without a wired link[18].



Fig. 23: Bluetooth logo.

d. Bluetooth Low Energy (IEEE 802.15.6)

Also known as Bluetooth Smart or BLE, it appeared in 2010 with the release of version 4.0 of the Bluetooth Core Specification. The Bluetooth Low Energy is an alternative to conventional Bluetooth, with a reduction in cost and power consumption while maintaining an equivalent range of communication[1].

e. Comparison

The table (**Tab. 6**) below represent a quick comparison between the previous wireless technologies [39]:

	Wi-Fi	ZigBee	Bluetooth	BLE
Speed (Mbps)	11 (b), 54 (a, g), 300 (n), 1300 (ac)[41].	0.25	0,7 (v1.2), 2.1 (v2.0), 2 (v2.1), 24 (v3.0- 4.1)[49].	0.1 [20].
Autonomy on a battery	Hours	Years	Days	Years [20].
Range (m)	300	10-100	10	10
Number of nodes	8	65000+	7	N/A

Tab. 6: Wireless technologies comparison.

From the table (**Tab. 6**) the best option to connect between microcontroller and the concentrator is ZigBee since it is energy efficient, with a good range, and we don't have a big amount of data. However between the concentrator and the gateway, Wi-Fi may be the best option for its high speed and range.

f. Data Structure

Due to the number of sensors and the presence name/value pairs, the sensors data will be sent from the microcontroller to the cloud in JSON format, however, the command will

be sent from the cloud to concentrator in JSON but from the concentrator to the microcontroller as a simple string in order to easily process it (see **Fig. 24**).

JSON: (JavaScript Object Notation) is a lightweight data-interchange format. It is a text format that is completely language independent but uses conventions that are familiar to programmers of the C-family of languages, including C, C++, C#, Java, JavaScript, Perl, Python, and many others. These properties make JSON an ideal data-interchange language, it is built on two structures [9]:

- A collection of name/value pairs. In various languages, this is realized as an object, record, struct, dictionary, hash table, keyed list, or associative array.
- An ordered list of values. In most languages, this is realized as an array, vector, list, or sequence.

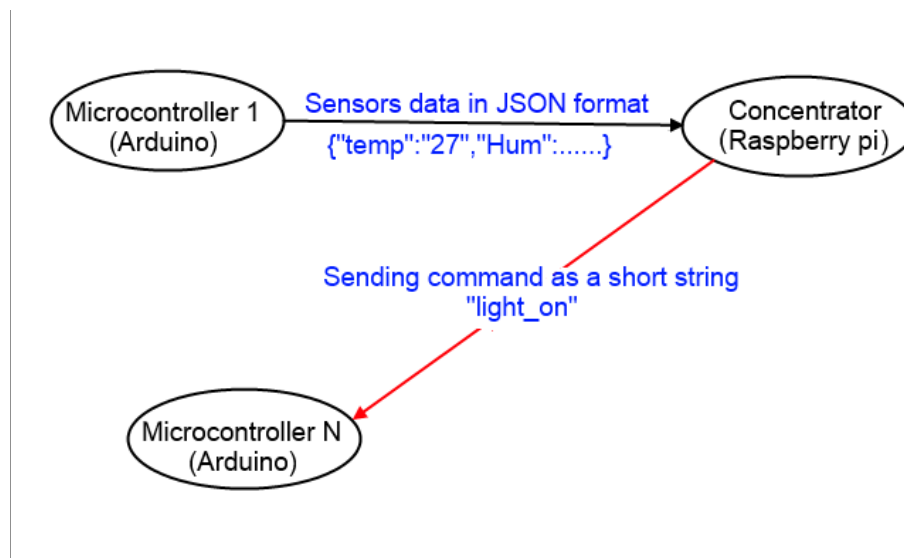


Fig. 24: Communication between the microcontrollers and the concentrator.

3.7. Implementing The Solution

This section will be divided into three parts, the first one represents the greenhouse (devices inside the greenhouse), the second is the gateway, and the last is IoT cloud.

3.7.1. The Greenhouse Side

A greenhouse designates a set of sensors, actuators, microcontrollers (Arduino Uno) and a concentrator (Raspberry Pi). Where the sensors and the actuators are connected to their microcontrollers, while the microcontrollers are connected to the concentrator.

a. Microcontrollers (Arduino Uno)

Every Greenhouse will have three microcontrollers, two of them are just for measurement and they're connected to the same set of sensors (**Fig. 25**), while the third one is connected to actuators (**Fig. 26**).

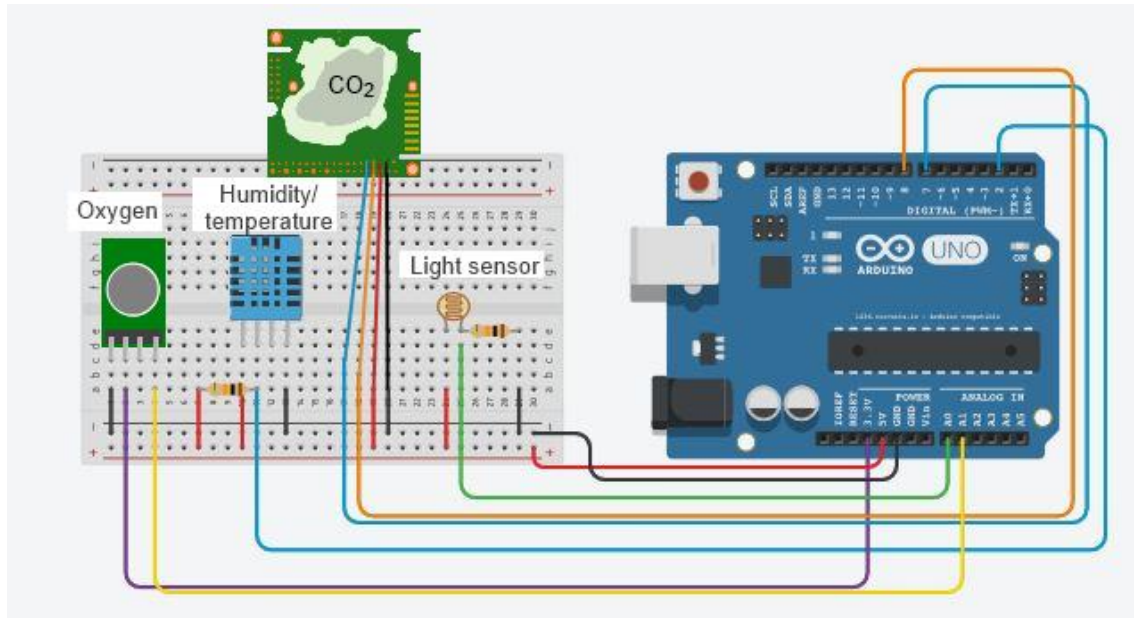


Fig. 25: Attaching the sensors.

The first two, collect sensors data, and send it to the raspberry pi in JSON format, their output result will be like:

```
{ "Temp": "23",
  "Hum": "55",
  "co2": "330",
  "Oxygen": "21",
  "Light": "1200" }
```

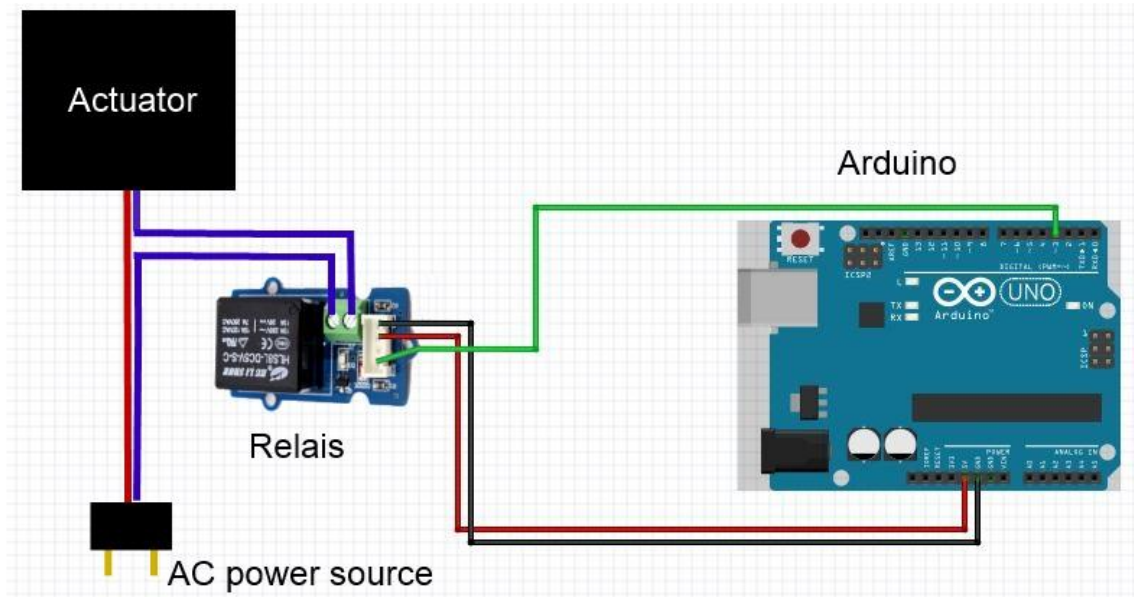


Fig. 26: Controlling an actuator with Arduino.

b. The Concentrator (The Raspberry Pi)

The following figure (**Fig. 27**) presents the Node-RED configuration of the concentrator (the Raspberry Pi of a greenhouse) and its function:

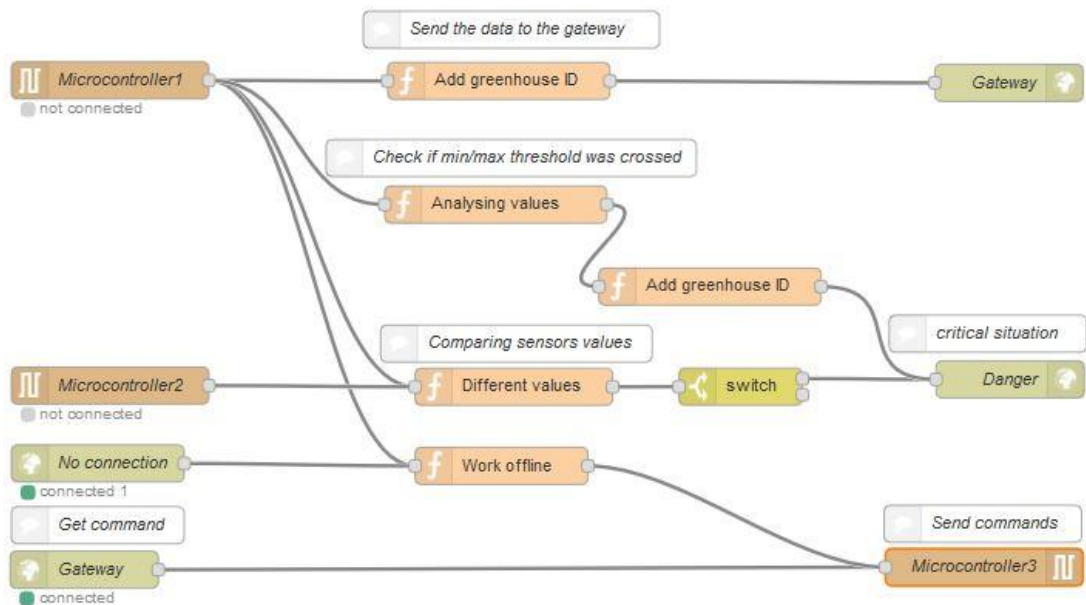


Fig. 27: Greenhouse Node-RED flows.

Both microcontrollers 1 and 2 have same sensors, while microcontroller 3 have only actuators.

Sensors data of the "Microcontroller 1" will be sent directly to the gateway in JSON format after adding the greenhouse identifier via the node "Add greenhouse ID" this will help the cloud recognizing which greenhouse sent this data, at the same time factors (temperature, light, humidity...) values are tested via the node "Analyzing values" so if one of them crossed the threshold an urgent message will be sent to the cloud through the gateway by the node "Danger" containing the JSON output of the first microcontroller and the greenhouse ID. Below is an example ("gh1" is the ID of the greenhouse 1):

```
{  "Danger": "1",
   "Topic": "gh1",
   "Payload": {
     "Temp": "23",
     "Hum": "55",
     "co2": "330",
     "Oxygen": "21",
     "Light": "1200"}
}
```

The flow doesn't only check if a factor value crossed its threshold, it also checks if sensors of both microcontrollers give same results, and if both microcontrollers are working, otherwise an urgent message will be sent to the cloud by the node "Danger" describing which sensor is not working properly, the message will look like:

```
{  "Danger": "1",
   "Topic": "gh1",
   "Payload": "Humidity sensors are not giving same result"
}
```

When the gateway can't connect to the cloud, it sends a message to the greenhouses using the "No connection" node, in this case, the greenhouses will work offline by processing the "microcontroller 1" sensors output and choosing actions, until connecting again to the IoT cloud where commands will be received from the cloud through the "Gateway" node.

3.7.2. The Gateway Side

The goal of the gateway is to link between the greenhouses and the IoT cloud, by sending sensors data to the cloud, getting commands from the cloud and sending it to the specified greenhouses. The flows below (**Fig. 28, 29, 30 and 31**) shows the gateway tasks.

a. Sending the Greenhouses Data to the Cloud

The following flow allows to send the greenhouses data to the cloud (**Fig. 28**):

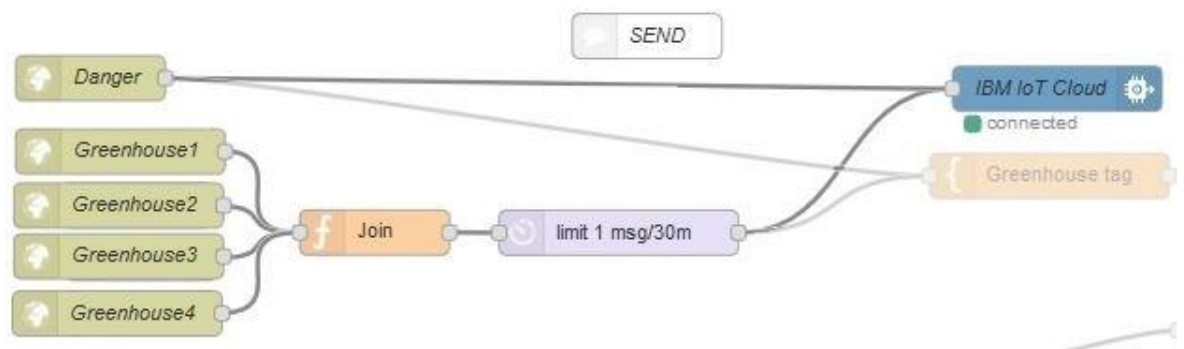


Fig. 28: Sending greenhouses data to the IoT cloud.

The “Danger” node represents the urgent message received from a greenhouse, this type of messages shouldn’t wait and have to be sent directly to the cloud.

The “Join” node will join the greenhouses messages (data) into one message represented as a JSON Array (each item of the array designate a greenhouse), and the “Limit” node sends to the cloud only one message every 30 minutes. Below is an example of the “Join” node output:

```
[
  {
    "Topic": "gh1",
    "Payload": {
      "Temp": "23",
      "Hum": "55",
      "co2": "330",
      "Oxygen": "21",
      "Light": "1200"
    }
  },
  {
    "Topic": "gh2",
    "Payload": { ... }
  },
  {
    "Topic": "gh3",
    "Payload": { ... }
  },
  {
    "Topic": "gh4",
    "Payload": { ... }
  }
]
```

b. Receiving Command From the Cloud

When receiving commands from the IoT cloud, the “switch” node, will send the command to the specified greenhouse (see **Fig. 29**):

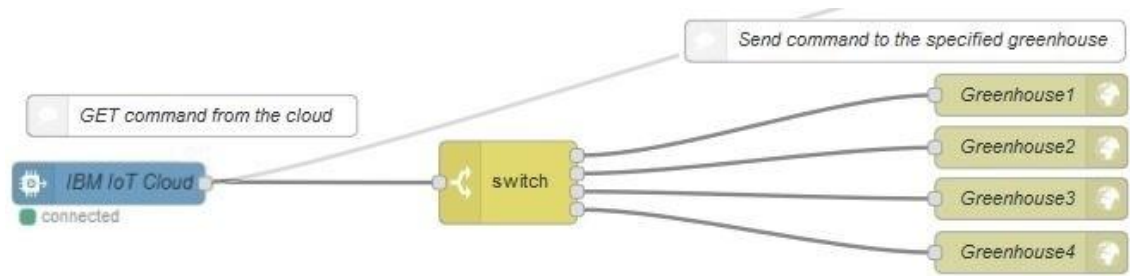


Fig. 29: Receiving the command from the IoT cloud and send it.

The received command is a JSON object, containing the greenhouse ID and the action name. If “gh1” is the greenhouse ID, and the “heateron” is the action name (to turn on the heater) the command will be like:

```
{"topic": "gh1", "payload": "heateron"}
```

c. Checking the Connection With the Cloud

If the gateway sends a message to the cloud(whether an urgent message via the “Danger” node or a normal via the “limit” node), it waits for a feedback from the cloud to confirm that the message was sent.

The concept here is to use the feedback to detect if the connection with the cloud was lost, if there’s no connection, the gateway will send the no connection command to the greenhouses to enable the offline mode, the following Node-RED flow represents the way used to detect this issue(see **Fig. 30**):

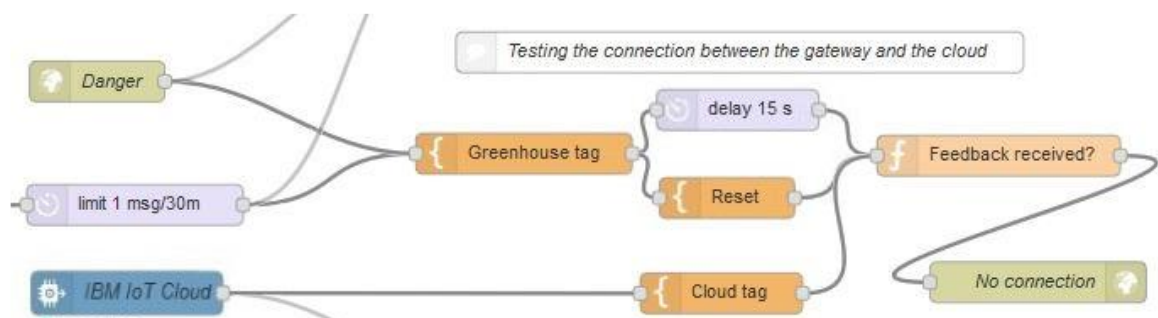


Fig. 30: Turning on the offline mode when losing the connection with the cloud.

Understanding the flow:

- The “Greenhouse tag” node replaces the message with the keyword “Greenhouse” indicating that the greenhouses sent a message to the cloud.

- The “delay” node will wait for 15 seconds before sending the “Greenhouse” keyword to the “Feedback received?” node.
- The “Cloud tag” node like “the greenhouse tag” replaces the message received from the cloud with the “Cloud” keyword, indicating that a message was received from the cloud.
- The “Feedback received?” function checks if the feedback was received from the cloud after sending a message within the delay of 15 sec. If no feedback received then the connection with the cloud was lost, and the gateway will order the greenhouses to work offline via the WebSocket node “No connection” (and vice-versa).
- The “reset” node, initialize the variables of the “Feedback received?” function.

d. The Complete Gateway Node-RED flow

Linking the previous flows together will result in the whole Node-RED flow of the gateway showed in the figure below (see **Fig. 31**):

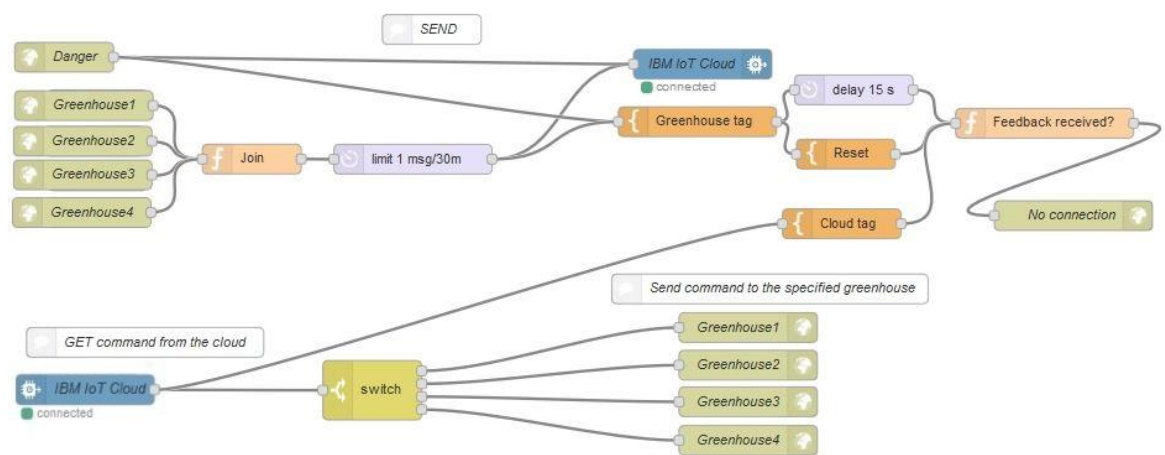


Fig. 31: The Complete Gateway Node-RED flow.

3.7.3. The Cloud Side

a. Analyzing and Storing Data

When the greenhouses data reach the IoT cloud, it will be stored in the cloud database which will help later in getting the data history, at the same time it will be processed to get the suitable action and then send it to the gateway (**Fig. 32**).

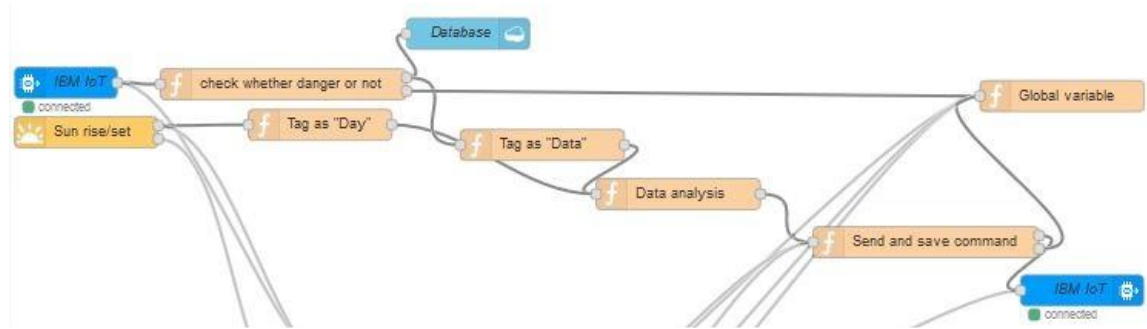


Fig. 32: IoT Cloud, Analyzing and storing greenhouses data.

The gateway message is received through the “IBM IoT” node, it can be a danger message indicating that something is wrong, or the greenhouses sensors data represented as a JSON array where each element of this array represents a greenhouse.

The message will be filtered via the function “Check whether danger or not”, if it is a danger message the greenhouses status will be updated (the global variable), else (in this case the message represents the greenhouses data) it will be processed and stored in the cloud database (via the “Database” node) for later use.

Before getting processed, the greenhouses data will be tagged as “Data” with the function “Tag as Data”, after that it will pass through the “Splitter” node which sends the array elements one by one to the “Data analysis” node.

The “Data analysis” node uses two inputs: the first is the greenhouse data which is tagged as “Data”, and because some factors thresholds changes from day to night the second input tells the function whether it is day or night. It also checks the weather information (**Fig. 38**) so if the weather outside the greenhouses matches the plant needs, the greenhouses will be opened (walls and roof) and the humidity and temperature related actuators will be turned off, which will help saving energy.

After processing, this function will choose the possible actions for every greenhouse and send them to the function “Send and save command” which as the name says, it sends the commands to the gateway through the “IBM IoT” output node, and saves it in the global variable.

b. Supplemental Lighting

The goal of the following flow is to calculate when the system should turn on or off the supplemental lighting (**Fig. 33**):

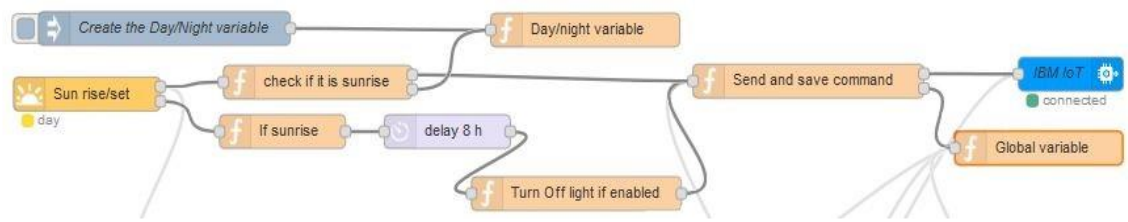


Fig. 33: IoT Cloud, supplemental lighting flow.

The “Create the day/night variable” will create the variable where the system stores day/night values and the sunrise time, this will allow to calculate the greenhouses exposition to the sun, and if the exposition time is inferior to the plant needs supplemental lighting will be activated.

The “if sunrise” function node, test if the sun rose, so it will start the timer (the delay node) which will automatically disable the supplemental lighting if it was already enabled (The 8 hours on the delay node, means that the plant needs about 8 hours of sunlight daily. So no need for supplemental lighting after the 8 hours ends).

The function “check if it is sunrise”, will check if it is a sunrise or a sunset, calculates the exposition time, and then decides if it should send turn on/off command of supplemental lighting.

c. Checking the Connection Between the Cloud and the Greenhouses

The following Node-RED flow (Fig. 34) aims to notify the user when the connection to the greenhouses is lost:

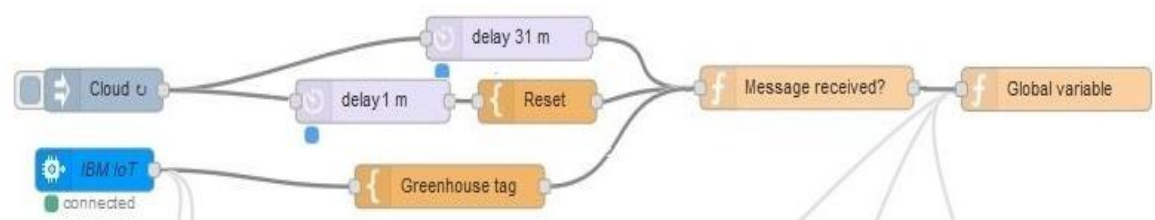


Fig. 34: Checking the connection between the IoT cloud and the greenhouses.

The greenhouses send their data to the cloud through the “IBM IoT node” every 30 min, so if no message was received after this delay, the cloud notifies the user that there is no connection between it and the greenhouses.

The “Message received” function needs three inputs:

- a. The input tagged as “Reset”: this will initialize the variables of this function, so it will start from the beginning.
- b. The input tagged as “Greenhouse”: if the function receive this input before receiving the “Cloud” called input, this means there is connection with the greenhouses.
- c. And The “Cloud” named input: when the function receives this as input, this means the delay of 31 min has finished. So if no greenhouse data was received in the last 31 min, the function will save the error message “No connection to the greenhouses” in the global variable to access it later via the mobile/web application. Otherwise, it will remove the message (if greenhouse data was received).

d. Remote Controlling

Remote controlling is the ability to perform a manual action (send a turn on/off command to a specified actuator of a greenhouse) from any location using the mobile/ web application.

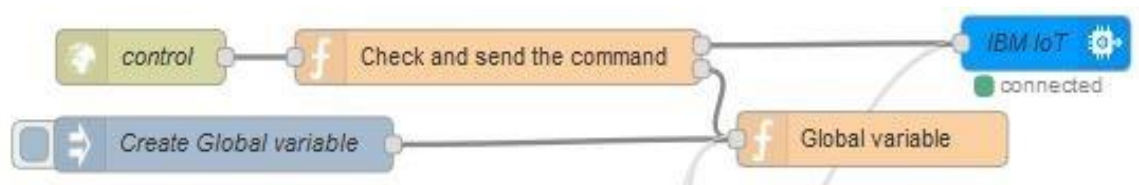


Fig. 35: Remote controlling flow.

The mobile/ web application sends the desired command to the cloud via the WebSocket protocol.

The function “Check and send the command” get the command from the WebSocket node “control”, analyses it, updates the global variable and sends the command to the gateway (see **Fig. 35**).

e. Getting the greenhouse status

Besides the remote controlling, the application provides the ability to see the greenhouses status (current values, error messages, and actuators states):

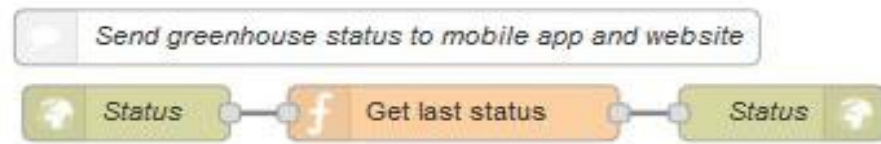


Fig. 36: Getting the greenhouse status.

The mobile/web application sends the request via the WebSocket protocol (the “Status” nodes, see **Fig. 36**), once the request received the “Get last status” function will read the global variable and sends the actuators states and the errors messages to the application.

f. Getting the history

The application should display the history of environmental factors values, the application sends the date of the requested values, and the cloud reads the database, searches (via the “Search” function) for the values matching that date, and then it sends them back to the application via the “Measures” WebSocket node (**Fig. 37**).



Fig. 37: Getting the greenhouses history.

g. Getting the Weather information

One of the Node-RED advantages, is the availability of many weather services nodes, in this project we used OpenWeatherMap service, to get weather information from the greenhouse location to the mobile and web applications. The figure below shows the Node-RED flows responsible of getting and sending the weather information (**Fig. 38**):



Fig. 38: Getting the weather and send it to the application.

h. Authentication

To ensure that only the specified user or users can access the data via the mobile or web application, the following flow (**Fig. 39**) will compare the email and the password entered by the user:



Fig. 39: Login Node-Red flow.

i. Getting the plant parameters

The flow below (**Fig. 40**) uses WebSocket protocol to get the plant information from the plant grow monitoring project mentioned early in the introduction, the “Create plant...” node initialize the variable declared in the function “Value of plant parameters”.

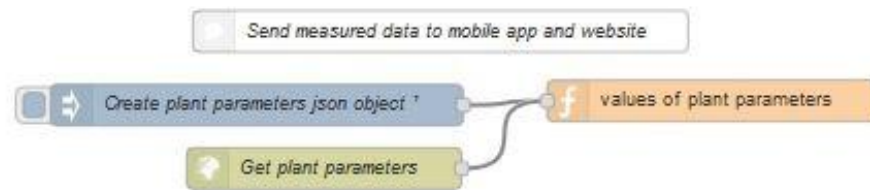


Fig. 40: Getting plant parameters values.

The plant parameters variable is a JSON

```
{ "temp": { "maxnight": "20", "maxday": "30", "minnight": "15", "minday": "20" },
  "hum": { "maxnight": "70", "maxday": "70", "minnight": "60", "minday": "60" },
  ... }
```

j. The complete Cloud Node-RED flow

Joining all the previous flows will result in the flowing Node-RED flow (**Fig. 41**):

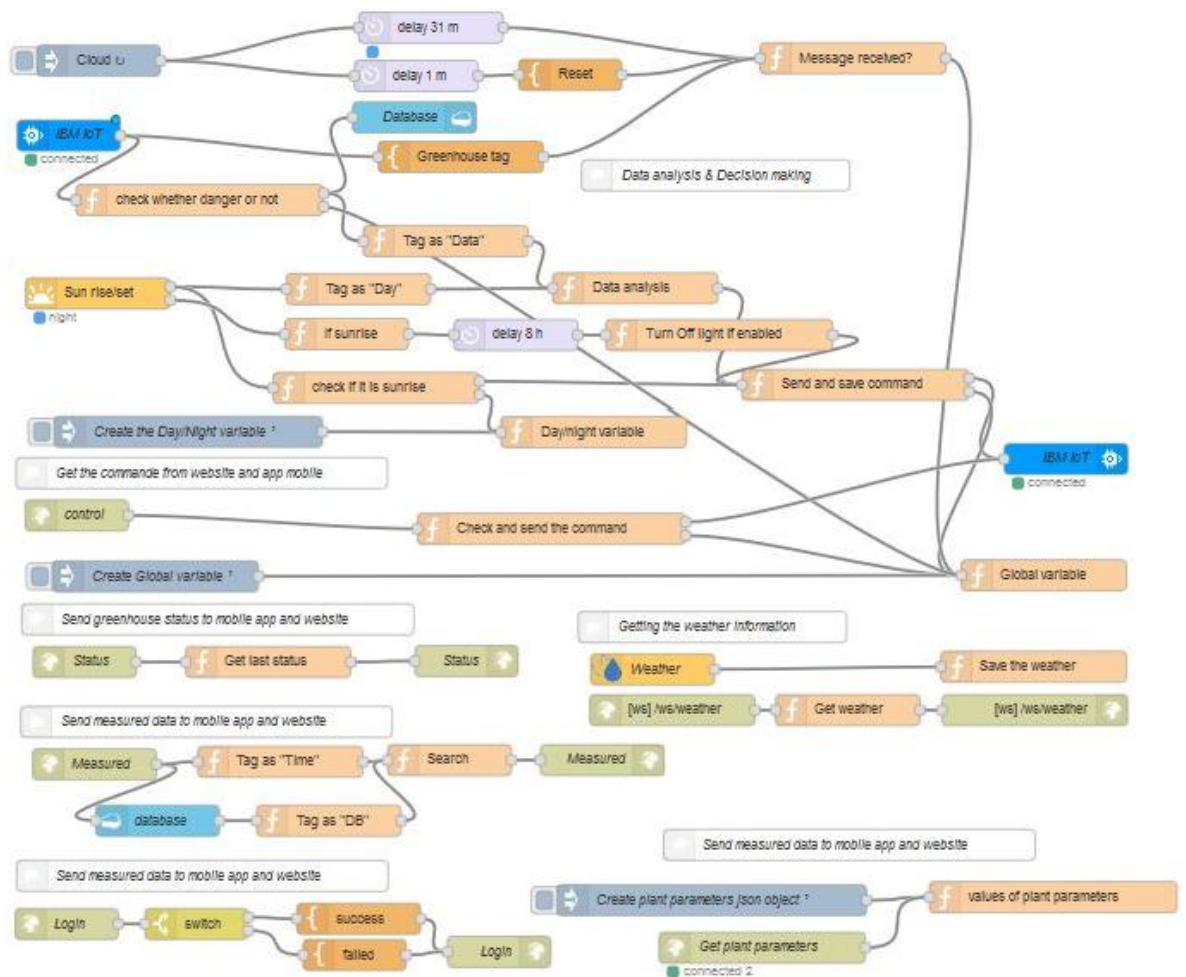


Fig. 41: The complete IoT Cloud Node-RED flow.

3.8. Mobile Application

3.8.1. Monitoring the Greenhouses

Once the application opened, the user will get in the greenhouse monitoring area (item 1, **Fig. 42**), in case there is an internet connection, and it will show directly the temperature history of that day of the “Greenhouse 1”.

The user can change which environmental factortodisplay (Item 3), of which greenhouse (item 2), and if he wanted to explore the previous days, he can choose the desired date (item 4).

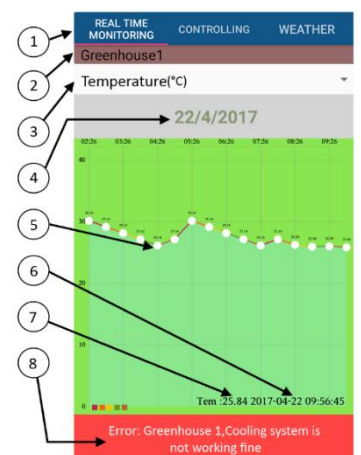


Fig. 42: The greenhouses monitoring tab.

The factors history is displayed as a chart, every point represents a value at a specific time (item 5), also in the bottom right corner of the chart, the application shows the last value of the selected factor and the time when it was sent to the cloud (items 6 and 7).

The application also shows the error message when something is not working fine (item 8), so the user can check the greenhouse.

3.8.2. Remote Controlling the Greenhouse

The application offers to the user the ability to execute actions manually, this can help when the farmer checks the greenhouses actuators for example, or when the weather outside is too good and the user wants to open the greenhouse. To do so, the application lists the available actuators (item 3 **Fig. 43**), and provides two separated options: manual mode (item 1) and on/off the actuator (item 5).

If the user wants to enable or disable an actuator, he must enable the manual mode of that actuator first (manual mode is enabled for the actuator “Fan», see item 2), and then toggle the on/off to enable or disable that actuator (item 4).

Once the manual mode is enabled, a notification will be shown in the notification drawer. So the user doesn’t forget it.

3.8.3. Weather Information

The weather tab, in the mobile application, represents the weather information of outside the greenhouses, the application shows a quick weather description (item 1 **Fig.44**), and the detailed weather information (item 2) which includes the outside temperature or humidity, the wind speed and its direction, and cloud coverage which means how much is that area covered with clouds (0% means a clear sky).

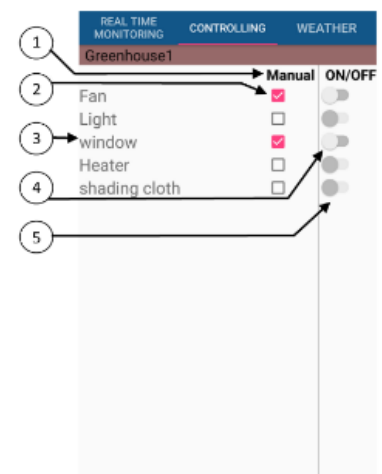


Fig. 43: Controlling the greenhouses via the mobile application.



Fig. 44: The weather information tab.

3.9. Web Application

The web application has the same options that the mobile application has, from choosing the greenhouse, the environmental factor and the date, to display the chart, to the manual control and the weather information (**Fig. 45**):

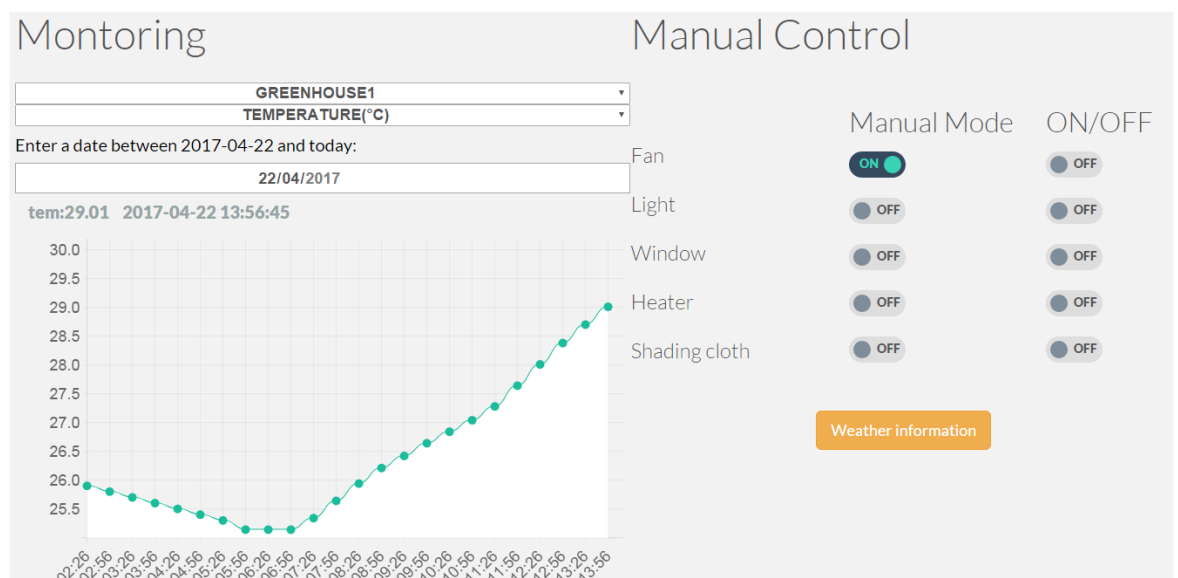


Fig. 45: The web application UI.

3.10. Prototyping

To test how the system perform in real life, we managed to build a prototype (**Fig. 46**).

Dimensions:

- Height: 115cm
- Width: 77cm
- Length: 102cm

Materials:

- Built with wood
- Covered with 1mm nylon sheet.

Actuators:

To test the system we used these actuators (**Fig. 47**):

- A led light bulb: as a lighting system
- PC Fans : as exhaust fans
- Removable walls
- Custom heater (consist of a 12v automotive light bulb Klaxcar H4 + a PC fan)



Fig. 46: The prototype.



Fig. 47: Some of the prototyping actuators.

3.11. Conclusion

In this part, we successfully implemented the solution of using Internet of things technology in monitoring and controlling the greenhouses environment.

We can't say that we didn't face any problem during this phase especially that we couldn't realize all the possibilities with the sensors and the actuators we have which forced us to only test one greenhouse with some actuators and simulate the other actuators and greenhouses. We also wish that we had all the project requirements and more time to test it in real conditions.

General Conclusion

In this project, we worked on agriculture, precisely in the greenhouse cultivation part. We successfully employed Internet of Things technology in the automation of the greenhouses and their remote monitoring with failures detection ability through mobile and web applications. It is important to mention that here we were interested only about the environmental factors and not about the plant growth which includes watering, treating diseases, and grow monitoring which is the second part of the project.

The IoT cloud based platform we have used allowed the free access to many cloud and web services (storage, processing, communication, notification...and so on) for example a weather service is used instead of using to replace a hardware weather station. We thought also about adding the video surveillance to this solution, but since we couldn't get all the sensors and actuators needed, we simulate them by injecting the values as if they were real sensors.

What's next? The project is only on its beginning and because "*Program testing can be used to show the presence of bugs, but never to show their absence!*." - Edsger W. Dijkstra [28], it still needs more testing and -depending on test results- it may needs improvement and bugs fixing.

To conclude, agriculture in general, is an extremely important field that we should care more about, with urbanization growing over green areas. Today, a lot of studies are focusing on increasing urban agriculture to benefit from wasted areas such as buildings roofs for growing food in order to face the urban and population growth, but as Dennis Dimick -*National Geographic's Executive Editor*- said: "Not only must we grow more food, we must grow more people interested in growing more food" [21] and this is why we think that an IoT based greenhouse system should be integrated in the smart building and smart city ecosystems.

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Abstract: Internet of Things is a new technology that allows objects, to interact and connect with one another through the internet, it gives many opportunities and can be used in many fields like in smart buildings, smart city, in health care, and more...

In this project we will use this technology in agriculture, precisely in greenhouse cultivation, which should result in automating the greenhouses, remote monitoring, controlling them, in order to maximize the production at a lower cost.

Keywords: Internet of things, greenhouse, automation, remote monitoring, control, cloud computing.

Système base sur l'IoT pour la télésurveillance et le control climatique des serres

Résumé : Internet des Objets, est une nouvelle technologie qui permet aux objets, d'interagir et de se connecter entre eux grâce à Internet, elle offre de nombreuses opportunités et peut être utilisé dans de nombreux domaines comme dans les bâtiments intelligents, la ville intelligente, le domaine de la santé et encore plus...

Dans ce projet, nous allons utiliser cette technologie dans l'agriculture, précisément dans l'agriculture sous serres, ce qui devrait nous permettre d'automatiser les serres, de les surveiller à distance, et de les contrôler, afin de maximiser la production à moindre coût.

Mots-clés: Internet des Objets, serre, automation, télésurveillance, contrôle, informatique en nuage.

نظام يعتمد على انترنت الأشياء من اجل المراقبة عن بعد والتحكم في مناخ البيوت البلاستيكية

ملخص: إنترنت الأشياء، هي تقنية جديدة تسمح للأشياء بالتفاعل والتواصل مع بعضها البعض من خلال شبكة الإنترنت، توفر العديد من الفرص، ويمكن استخدامها في العديد من المجالات مثل المباني الذكية، المدن الذكية، مجال الصحة والعديد من المجالات الأخرى...

في هذا المشروع، سوف نستخدم هذه التكنولوجيا في مجال الزراعة تحديدا في البيوت البلاستيكية، والتي ينبغي أن تسمح لنا بالتشغيل الآلي للبيوت البلاستيكية، مراقبتها عن بعد والتحكم بها، لتعزيز الإنتاج بأقل تكلفة.

الكلمات الدالة: إنترنت الأشياء، بيوت بلاستيكية، تشغيل آلي، مراقبة عن بعد، تحكم، حوسبة سحابية.